

Tutorial 7: Blockchain Mechanics and the Double-Spending Problem

Financial Economics
Research School of Economics, Australian National University
Instructor: Simon Mishricky

Source. The problems below are based on Chiu and Koepl (2017), *The Economics of Cryptocurrencies – Bitcoin and Beyond* (cited throughout as CK2017).

Learning objectives. After completing this tutorial you should be able to

- explain how a blockchain functions as a decentralised record-keeping device and how Proof-of-Work (PoW) underpins consensus;
- articulate the double-spending problem and the role of confirmation lags;
- analyse the mining game and derive the no-double-spending constraint;
- compute the probability of a successful secret-mining attack as a function of trade size, rewards, and confirmation lag.

1. **Blockchain as memory.** A central claim of CK2017 is that the blockchain is a record-keeping device.

- (a) In one or two paragraphs, contrast a blockchain with traditional fiat money in terms of how each “remembers” past transactions. Why does this difference matter for decentralised payments?
- (b) Why does the longest-chain rule make rewriting old transactions costly? Explain in terms of the PoW that must be redone if a malicious miner wants to revise a block k periods in the past.
- (c) Explain in your own words why a blockchain alone does *not* eliminate forward-looking double-spending attacks.

2. **The mining game.** Following Section 3.2 of CK2017, M symmetric miners each choose computing power $q(i)$ at unit cost. The probability that miner i wins block n is

$$\rho(i) = \frac{q(i)}{\sum_{m=1}^M q(m)}.$$

The winner receives reward R (in real balances), discounted by β across the period.

- (a) Write down the miner’s per-subperiod problem and derive the first-order condition for $q(i)$.

- (b) Imposing symmetry $q(m) = Q$, show that the Nash equilibrium choice is

$$Q = \frac{M-1}{M^2} \beta R.$$

- (c) Compute the aggregate computing cost MQ and show that as $M \rightarrow \infty$, $MQ \rightarrow \beta R$. Interpret this “rent dissipation” result.
- (d) Compute the expected profit of each miner across the full period of $\bar{N} + 1$ subperiods and show that it vanishes in the competitive limit.

3. **Double-spending attack mechanics.** Consider the secret-mining attack of Section 3.3 of CK2017. The buyer must mine $N + 1$ blocks faster than the rest of the network. Let $\Delta = d/R + (N + 1)$.

- (a) Explain the intuition for why the buyer must mine $N + 1$ *consecutive* blocks (not N).
- (b) In the final round (subperiod N), the buyer’s payoff from investing q_N is

$$\rho(q_N) \beta [d + (N + 1)R] - q_N,$$

where the second term in brackets reflects the $N + 1$ block rewards the buyer would claim after broadcasting his secret chain. Take $M \rightarrow \infty$. Show that the optimal investment satisfies

$$\hat{q}_N(d, N) = \beta R [\sqrt{\Delta} - 1]$$

and that the corresponding success probability is

$$\rho(\hat{q}_N) = \frac{\sqrt{\Delta} - 1}{\sqrt{\Delta}}.$$

- (c) *Backward induction.* Using the recursive structure $D_n(d, N) = \max_{q_n} \rho(q_n) D_{n+1}(d, N) - q_n$ with terminal condition $D_N(d, N) = \beta R (\sqrt{\Delta} - 1)^2$, verify by induction that

$$D_{N-s}(d, N) = \beta R [\sqrt{\Delta} - (s + 1)]^2, \quad s = 0, 1, \dots, N.$$

4. **The no-DS constraint.**

- (a) Use the result $D_0(d, N) = \beta R [\sqrt{\Delta} - (N + 1)]^2$ to derive the “double-spending proof” condition $d < R(N + 1)N$.
- (b) Show that the unconditional probability of a successful double spend (when the DS constraint fails) is

$$P(d, N) = \frac{\sqrt{\Delta} - (N + 1)}{\sqrt{\Delta}}.$$

- (c) Discuss qualitatively how $P(d, N)$ varies with d , R , and N . In particular, explain why doubling the confirmation lag has a non-linear effect on attack probability.
- (d) Suppose $R = 0.5$ and consider $N = 6$ (Bitcoin’s typical merchant standard). What is the largest payment d that can be settled with full finality? Now repeat for $N = 1$. Comment on the trade-off.

5. **Immediate vs. final settlement.** CK2017 prove a sharp impossibility theorem: under any PoW protocol, settlement cannot be both immediate and final.

- (a) Show, using your derivations above, that at $N = 0$ the DS constraint $d < R(N + 1)N$ fails for any $d > 0$.
 - (b) Explain in plain English what this implies for retail spot trade in a Bitcoin-like system.
 - (c) Compare and contrast the cryptocurrency trade-off (security vs. speed of settlement) with the trade-off faced by a centralised payment system (e.g. Visa, Fedwire).
6. **Mining-reward composition (short).** The mining reward in the general equilibrium model is

$$R = \frac{Z(\mu - 1) + D\tau}{\bar{N} + 1},$$

where μ is the gross money growth rate, τ the transaction fee, Z aggregate balances, and D aggregate transfers.

- (a) Explain the economic interpretation of each of the two terms in the numerator.
- (b) Why is R divided by $\bar{N} + 1$?
- (c) Discuss why, holding R fixed, the choice between financing rewards via μ or τ might affect different buyers asymmetrically. (We will return to this in Tutorial 8.)

References

- [1] Chiu, J. and T. V. Koepl (2017). The Economics of Cryptocurrencies – Bitcoin and Beyond. *Bank of Canada Staff Working Paper / Queen's University*.
- [2] Lagos, R. and R. Wright (2005). A Unified Framework for Monetary Theory and Policy Analysis. *Journal of Political Economy* 113(3), 463–484.